

FLORENCE, SC, USA

WMO: 723106

Lat: 34.188N

Lon: 79.731W

Elev: 146

StdP: 14.62

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13744

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.6	27.0	5.6	7.4	32.7	10.3	9.4	35.4	23.7	54.3	20.4	53.4	4.9	340	0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	96.0	76.5	93.6	75.9	91.5	75.5	79.3	90.4	78.3	88.7	77.5	87.3	8.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.5	139.1	83.1	75.5	134.4	82.2	74.9	131.6	81.6	42.8	90.2	41.8	89.1	40.9	87.3	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	17.5	15.5	15.5	99.2	4.6	3.0	12.2	101.3	9.5	103.1	7.0	104.8	3.7	107.0
			13.8	81.3	4.2	1.6	10.8	82.5	8.4	83.4	6.0	84.3	3.0	85.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.4	45.9	49.2	55.8	64.1	71.9	78.8	81.5	80.3	75.2	65.3	54.8	48.9	(6)
	DBStd	14.64	10.45	9.54	9.06	7.49	6.31	4.69	3.58	3.74	5.28	7.47	8.43	9.56	(7)
	HDD50	565	207	122	48	3	0	0	0	0	0	2	44	139	(8)
	HDD65	2392	594	448	306	107	17	1	0	0	0	90	320	505	(9)
	CDD50	5815	80	100	228	426	678	863	975	940	755	476	189	105	(10)
	CDD65	2168	3	6	21	80	230	414	510	475	309	99	15	6	(11)
	CDH74	20333	12	47	220	799	2124	4001	5240	4550	2467	740	101	32	(12)
	CDH80	8508	0	4	34	210	794	1819	2483	2051	933	170	8	2	(13)
(14) Wind	WSAvg	6.9	7.4	7.8	8.2	8.2	7.3	6.5	6.2	5.7	6.5	6.2	6.3	6.7	(14)
(15) Precipitation	PrecAvg	43.2	3.6	3.1	3.9	2.7	3.2	4.2	5.6	5.0	3.5	2.6	2.4	2.9	(15)
	PrecMax	59.0	7.0	6.9	10.9	6.1	8.2	9.3	11.1	14.4	8.8	10.8	8.2	6.5	(16)
	PrecMin	29.4	0.7	0.8	0.4	0.2	0.5	0.6	1.5	1.5	0.0	0.0	0.4	0.4	(17)
	PrecStd	7.3	1.8	1.6	2.2	1.6	1.7	2.1	2.4	2.9	2.2	2.4	1.7	1.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.7	79.9	83.8	88.9	94.8	98.0	98.9	99.1	93.6	87.9	80.9	78.2	(19)
		MCWB	64.1	64.9	64.9	68.5	73.2	76.7	77.3	77.1	73.9	72.9	67.8	65.4	(20)
	2%	DB	71.3	74.4	80.0	85.3	90.5	94.9	95.9	94.7	90.9	84.2	77.1	73.2	(21)
		MCWB	62.1	62.0	63.5	67.3	71.7	75.7	77.2	76.8	74.2	70.4	65.0	64.2	(22)
	5%	DB	67.1	70.1	75.9	82.3	87.8	92.2	93.4	92.2	88.5	81.4	73.4	69.1	(23)
		MCWB	59.6	59.0	61.1	65.5	70.9	75.0	76.8	76.3	73.5	68.1	63.1	63.0	(24)
	10%	DB	63.0	65.6	72.0	79.0	84.6	89.5	91.0	89.9	85.8	78.3	69.7	65.0	(25)
		MCWB	56.9	56.2	59.1	64.1	69.5	74.1	76.4	75.8	72.7	67.6	62.0	59.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.4	67.8	68.5	72.8	76.2	79.7	81.0	80.3	78.3	77.2	71.0	69.6	(27)
		MCDB	71.3	75.2	77.0	81.5	87.1	93.0	92.4	90.9	86.7	81.0	74.8	73.6	(28)
	2%	WB	64.3	65.0	66.3	70.4	74.4	78.1	79.6	79.1	77.1	74.6	68.5	66.8	(29)
		MCDB	69.3	71.9	74.6	79.8	84.9	89.8	91.0	89.3	85.4	80.2	73.5	70.6	(30)
	5%	WB	61.3	61.9	64.3	68.6	73.1	77.0	78.4	78.2	76.0	72.3	66.2	63.8	(31)
		MCDB	65.8	67.6	72.0	77.6	83.0	87.6	89.1	87.9	83.5	77.8	71.2	68.0	(32)
	10%	WB	57.9	58.6	62.1	66.8	71.7	75.9	77.7	77.3	74.9	70.0	63.5	60.5	(33)
		MCDB	62.0	64.3	69.3	75.2	80.7	85.4	87.8	86.5	81.7	75.8	68.5	65.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	20.3	21.4	22.8	23.4	21.5	19.2	18.5	18.1	18.4	21.0	22.5	20.1	(35)
		MCDBR	24.2	26.1	26.9	26.3	24.2	22.2	21.4	21.0	21.6	23.3	25.1	23.3	(36)
	5% WB	MCWBR	16.0	16.0	13.2	11.2	8.7	7.1	6.8	6.6	7.8	10.7	14.6	15.4	(37)
		MCWBR	20.6	24.0	22.4	21.7	20.2	19.4	19.1	18.4	17.3	17.1	20.4	20.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.318	0.327	0.361	0.405	0.467	0.493	0.542	0.513	0.430	0.392	0.341	0.316	(40)
		taud	2.501	2.472	2.406	2.270	2.152	2.123	2.006	2.078	2.299	2.374	2.475	2.545	(41)
	Ebn at Noon		283	292	289	280	263	255	241	246	264	265	272	278	(42)
		Edh at Noon	27	30	36	43	49	50	56	51	39	34	27	24	(43)
(44) All-Sky Solar Radiation	RadAvg		871	1074	1426	1795	1950	2029	1936	1750	1501	1248	958	764	(44)
	RadStd		48	113	114	123	154	139	120	84	126	148	77	64	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+338	+266
(47)	Regional (62 neighbors)	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+320	+232

Nomenclature: See separate page